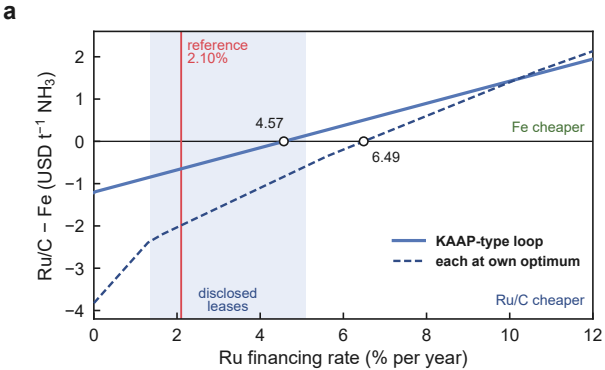
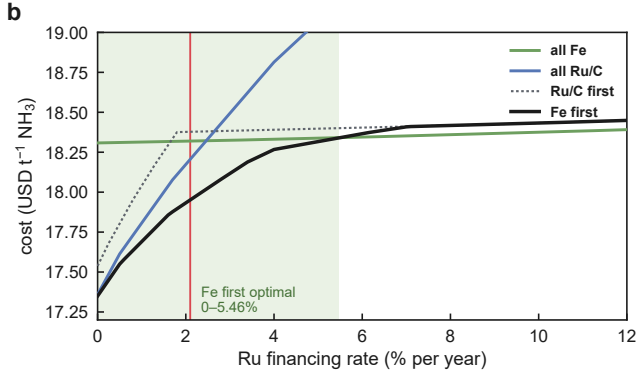
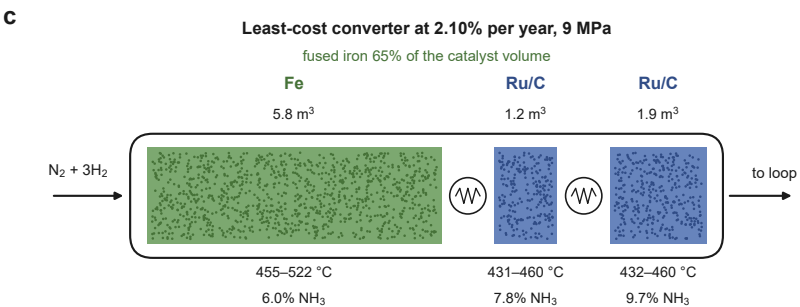
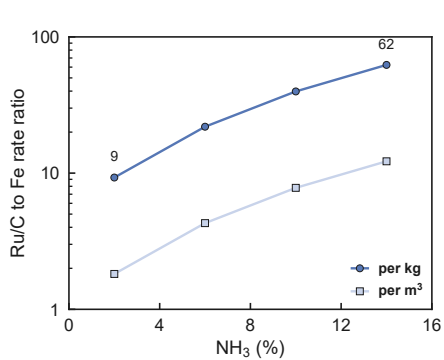

a
reference
2.10%
4.57
6.49
Fe cheaper
Ru/C cheaper
KAAP-type loop
each at own optimum
disclosed
leases
Ru/C – Fe (USD t^{-1} NH$_3$)
Ru financing rate (% per year)


b
all Fe
all Ru/C
Ru/C first
Fe first
Fe first optimal
0–5.46%
cost (USD t^{-1} NH$_3$)
Ru financing rate (% per year)


c
Least-cost converter at 2.10% per year, 9 MPa
fused iron 65% of the catalyst volume
Fe
Ru/C
Ru/C
5.8 m^3
1.2 m^3
1.9 m^3
N$_2$ + 3H$_2$
to loop
455–522 °C
6.0% NH$_3$
431–460 °C
7.8% NH$_3$
432–460 °C
9.7% NH$_3$


d
9
62
per kg
per m^3
Ru/C to Fe rate ratio
NH$_3$ (%)